Second-order gain model: observability matrix

1 Model

State equation

$$\delta x_1[k+1] = \delta x_2[k]$$
$$\delta x_2[k+1] = \delta x_3[k]$$
$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] + dF_{dx}[k] \cdot e_{\delta a_x}[k] - \varepsilon[k]$$
$$a_x[k+1] = a_x[k] + \delta a_x[k]$$
$$\delta a_x[k+1] = \delta a_x[k]$$

$$F_{dx} \approx \big(1 + (1-\lambda_c)d\big)\,\frac{\Delta x_d}{\hat{a}_x}, \qquad dF_{dx} = (1-\lambda_c)\sum_{i=1}^{d} i \cdot f_{dx}[k-i] \qquad \text{(constant velocity)}$$

Output equation

$$y_1[k] = \delta x_1[k] + n_x[k], \qquad R_1 = \mathrm{Var}(n_x)$$
$$y_2[k] = a_x[k-d] + n_a[k] \approx a_x[k] - d \cdot \delta a_x[k] + n_a[k], \qquad R_2 = \mathrm{Var}(n_a)$$

Estimation error and output error

$$\mathbf{e}[k] = \begin{bmatrix} e_{\delta x_1} & e_{\delta x_2} & e_{\delta x_3} & e_{a_x} & e_{\delta a_x} \end{bmatrix}^T [k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = e_{\delta x_1}[k] + n_x[k]$$
$$e_{y_2}[k] = y_2[k] - \hat{y}_2[k] = e_{a_x}[k] - d \cdot e_{\delta a_x}[k] + n_a[k]$$

Error propagation (no noise)

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k], \quad \mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] & dF_{dx}[k] \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} e_{y_1}[k] \\ e_{y_2}[k] \end{bmatrix} = \mathbf{H}\,\mathbf{e}[k], \quad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -d \end{bmatrix}$$

2 Observability matrix

Errors at step k_0, output errors j steps later

$$\begin{bmatrix} e_{y_1}[k_0+j] \\ e_{y_2}[k_0+j] \end{bmatrix} = \mathbf{H}\,\mathbf{F}_e[k_0+j-1]\cdots\mathbf{F}_e[k_0]\;\mathbf{e}[k_0]$$

The gain block moves as a line

$$e_{a_x}[k_0+j] = e_{a_x}[k_0] + j\cdot e_{\delta a_x}[k_0], \qquad e_{\delta a_x}[k_0+j] = e_{\delta a_x}[k_0]$$

The matrix (columns $e_{\delta x_1}, e_{\delta x_2}, e_{\delta x_3}, e_{a_x}, e_{\delta a_x}$ at k_0)

$$\begin{aligned}
y_2,\ j\geq 0: &\quad [\,0\ \ 0\ \ 0\ \ 1\ \ j-d\,] \\
y_1,\ j=0,1,2: &\quad [\,1\,0\,0\,0\,0\,],\ \ [\,0\,1\,0\,0\,0\,],\ \ [\,0\,0\,1\,0\,0\,] \\
y_1,\ j\geq 3: &\quad \Big[\,0\ \ 0\ \ \lambda_c^{j-2}\ \ -\textstyle\sum_{i=0}^{j-3}\lambda_c^{j-3-i}F_{dx}[k_0+i]\ \ \sum_{i=0}^{j-3}\lambda_c^{j-3-i}\big(dF_{dx}[k_0+i]-i\,F_{dx}[k_0+i]\big)\,\Big]
\end{aligned}$$

The gain columns (F_{dx} constant, j large)

$$e_{y_2}[k_0+j]\Big|_{\text{gain}} = 1\cdot\big(e_{a_x}[k_0]+(j-d)\,e_{\delta a_x}[k_0]\big)$$

$$e_{y_1}[k_0+j]\Big|_{\text{gain}} \approx -\frac{F_{dx}}{1-\lambda_c}\big(e_{a_x}[k_0]+(j-j_1)\,e_{\delta a_x}[k_0]\big), \qquad j_1 = 3+\frac{\lambda_c}{1-\lambda_c}+\frac{dF_{dx}}{F_{dx}}$$

3 What the matrix shows

Every row reads the gain error at one moment on a straight line

$$e_{a_x}[k_0]+\tau\,e_{\delta a_x}[k_0] = e_{a_x}[k_0+\tau], \qquad \tau = j-d\ (y_2), \qquad \tau = j-j_1\ (y_1)$$

Speed enters only the y_1 factor

$$y_2\text{ factor} = 1, \qquad y_1\text{ factor} = -\frac{F_{dx}}{1-\lambda_c}\propto\frac{\Delta x_d}{\hat{a}_x}, \qquad \Delta x_d = \frac{(h_0-h_b)\,R}{T_d/T_s}$$

4 Solving the matrix equation: weighted line fit

4.1 Each row as a reading of the gain at time $k_0 + \tau_j$ (y_1 rows divided by their factor)

$$z_j = e_{a_x}[k_0] + \tau_j\, e_{\delta a_x}[k_0] + \text{noise}_j, \qquad w_j = \frac{1}{\text{Var}(\text{noise}_j)} = \frac{1}{R_2}\ (y_2), \quad \frac{1}{R_1}\Big(\frac{F_{dx}}{1-\lambda_c}\Big)^2\ (y_1)$$

4.2 Two unknowns that fit every row best

$$\text{minimize} \quad \sum_j w_j \big(z_j - e_{a_x}[k_0] - \tau_j\, e_{\delta a_x}[k_0]\big)^2$$

$$\frac{\partial}{\partial e_{a_x}},\ \frac{\partial}{\partial e_{\delta a_x}} = 0: \quad \begin{bmatrix} \sum_j w_j & \sum_j w_j\,\tau_j \\ \sum_j w_j\,\tau_j & \sum_j w_j\,\tau_j^2 \end{bmatrix} \begin{bmatrix} \hat{e}_{a_x}[k_0] \\ \hat{e}_{\delta a_x}[k_0] \end{bmatrix} = \begin{bmatrix} \sum_j w_j\, z_j \\ \sum_j w_j\,\tau_j\, z_j \end{bmatrix}$$

4.3 Back to the gain: a straight line through the true gains

$$z_j = a_x[k_0 + \tau_j] - \big(\hat{a}_x[k_0] + \tau_j\, \delta\hat{a}_x[k_0]\big) + \text{noise}_j$$

$$\hat{a}_x[k_0 + N] = A + B\,N, \qquad (A, B) = \arg\min_{A,B} \sum_j w_j \big(a_x[k_0 + \tau_j] - A - B\,\tau_j\big)^2$$

$$e_{a_x}[k_0 + N] = a_x[k_0 + N] - \big(A + B\,N\big)$$

5 T_d

5.1 Constant velocity: step number and height are the same axis

$$h_d[k_0 + \tau] = h_0 - \tau\,\Delta h_d, \qquad \Delta h_d = \frac{(h_0 - h_b)\,T_s}{T_d} \quad \Rightarrow \quad A + B\,\tau = A + \frac{B}{\Delta h_d}\big(h_0 - h_d\big)$$

5.2 The same fit written over height

$$\sum_j w_j \big(a_x(h_d[j]) - A' - B'\,h_d[j]\big)^2 = \sum_{\text{bands}} \big(\text{steps in band}\big) \cdot \big(w \text{ per step}\big) \cdot \big(a_x - A' - B'\,h_d\big)^2$$

$$\text{steps in band} \propto T_d \text{ (every band)}, \qquad w \text{ per step} = \underbrace{\frac{1}{R_2}}_{\text{no } T_d} + \underbrace{\frac{1}{R_1}\Big(\frac{F_{dx}}{1-\lambda_c}\Big)^2}_{\propto 1/T_d^2}$$

5.3 Result

y_2 only: objective $\propto T_d \ \Rightarrow\ (A', B')$ and $\hat{a}_x$ at h_b independent of T_d

y_1 added: weight near the wall $\propto 1/T_d \ \Rightarrow$ line pulled less towards the wall as T_d grows

6 Kalman filter

Gain block, $\mathbf{Q} = 0$, information form

$$\mathbf{F}_a = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad \left(\mathbf{P}^f[k]\right)^{-1} = \mathbf{F}_a^{-T}\,\mathbf{P}^{-1}[k-1]\,\mathbf{F}_a^{-1} \qquad \text{(old rows moved to step } k\text{)}$$

$$\mathbf{P}^{-1}[k] = \left(\mathbf{P}^f[k]\right)^{-1} + w_k \begin{bmatrix} 1 \\ -d \end{bmatrix} \begin{bmatrix} 1 & -d \end{bmatrix} \qquad \text{(this step's row added)}$$

$\mathbf{P}^{-1}[k]$ = the 2×2 matrix of 4.2, all rows up to step k $\Rightarrow$ $\hat{a}_x[k]$ = the line of 4.3 at step k

Descent only, $T_d = 5$ s vs 50 s: filter $\hat{a}_x$ (blue) vs the weighted line fit of 4.3 (green dashed)

